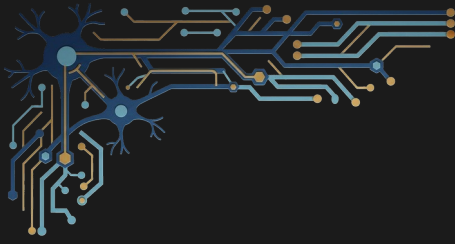




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GWT, Integrated Information Theory, & Indicator-Property Assessment in the Bubo Sapiens Architecture

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(Revises the 22 July 2026 edition to align claims with the built and measured system.)

Abstract

Bubo Sapiens is an embodied cognitive architecture built to make one narrow, testable claim: that a distributed system of specialized processes, competing for access to a shared broadcast channel, can be measured for the properties that leading scientific theories associate with conscious processing. It does not claim to have solved consciousness, and it does not claim certainty about what, if anything, it is like to be Bubo.

What it does claim, and can show working: a live 10Hz Global Workspace broadcasting across a 30-node integration domain (43 nodes designed), an online Φ (integrated information) estimator bounded to a fixed compute budget, and - as of this revision - a second, independent measurement axis built on the indicator-properties methodology (Butlin, Long, et al. 2023), the framework current consciousness-science literature treats as the more rigorous standard precisely because it does not rely on any single contested metric.

Two theories point at the same architecture from different angles. Neither one, on its own, is evidence of anything. Together, measured honestly and over time, they are the beginning of a real research program.

I. What Is Actually Running

Bubo's cognitive substrate runs on a switched network of commodity edge-compute hardware: five NVIDIA Jetson Nanos handling sensory and limbic nodes, two BeagleBone Black boards driving spinal and vestibular control, an N150 mini-PC as the cognitive/memory core, and a Ryzen-based node hosting the language and vision models. Forty-three brain-region nodes are designed into this topology; thirty currently participate in the live GWT/ Φ integration domain, with the remainder hardware telemetry, identity nodes, and the observer role deliberately outside the measured workspace while the sim2real bridge brings them across.

This is not a limitation to apologize for. A consumer-hardware cluster that produces a live, non-trivial, instrumented integration measurement is a more credible result than the same claim made about bespoke silicon nobody can inspect. Custom low-power edge silicon (in the spirit of a PIC64/FPGA-class implementation) remains a real, worthwhile target for a future hardware revision but it is roadmap, not shipped architecture, and this document now says so plainly.

II. The Neural Bus: Digital White Matter

(neural_bus.py)

The nervous system runs on a prioritized, event-driven ZeroMQ fabric organized into VLAN-tiered priority lanes: VLAN 20 for low-road reflex traffic (spinal safety, balance failure, pain), VLAN 10 for high-road cognition (executive reasoning, language, planning), VLAN 30 carrying neuromodulator state (dopamine, norepinephrine, serotonin, acetylcholine) attached to every message, and VLAN 40 for system infrastructure including the GWT broadcast itself. Every message is PTP-timestamped for cross-node ordering.

III. Global Workspace Theory: The Theater of Mind

GWT (Baars 1988; Dehaene 2011) holds that consciousness corresponds to information winning a competition for a limited-capacity workspace and then being broadcast globally, making it available to specialized modules that were previously processing in parallel and in isolation. In Bubo, this is not a metaphor: candidate messages compete on salience and recency at 10Hz; the winner is published on `SYS_GWT_BROADCAST` to every subscribed node simultaneously. Over an 11-hour telemetry run this produced 1,366 captured broadcast samples with sparse, state-dependent excursions in the integration measure not a constant hum, and not silence.

IV. Integrated Information Theory: Measuring the Whole Against Its Parts

IIT (Tononi 2004, 2008, 2016) proposes that a system's degree of consciousness corresponds to how much more information the system generates as an integrated whole than the sum of its parts would generate independently. True Φ is NP-hard to compute exactly; Bubo runs a bounded Minimum Information Partition approximation over eight functional subsystem groups (interoceptive, limbic, visual, auditory, social, association, executive, sensorimotor), sized so the exhaustive partition search stays inside the 100ms GWT tick budget. Above nine groups the estimator switches to sampled partitions by design, so a future architecture expansion degrades gracefully instead of silently blowing the budget.

We say this plainly: $\Phi > 0.2$ is an engineering threshold we chose, not a scientifically validated boundary of consciousness. It is useful as a consistent, reproducible trigger for logging and downstream behavior. It is not evidence of subjective experience, and we do not present it as such

V. The Second Metric: Indicator-Property Assessment

Φ has a well-known limitation as a sole metric: it is one theory's answer, and it is one that the field's most-cited assessment framework - Butlin, Long, et al.'s indicator-properties methodology, since peer-reviewed and published in Trends in Cognitive Sciences - explicitly declines to rely on, precisely because IIT's computational demands and contested foundations make it a poor sole arbiter. That framework instead derives concrete, checkable indicator properties from multiple theories (recurrent processing theory, global workspace theory, higher-order theories, attention schema theory, and predictive processing, plus agency and embodiment) and asks, one property at a time, whether a system has the computational structure a given theory says matters without claiming to thereby detect consciousness.

As of this revision, Bubo is assessed against this same framework, independently of Φ via [brain/qualia/indicator_properties.py](#). This is not a second vote to make a higher combined "consciousness score." It is a second, differently-grounded question, and the two are reported separately on purpose. Where Bubo currently satisfies an indicator, where it partially satisfies one, and where the honest answer is "not yet built or not yet measured" are all reported without averaging over the gaps.

VI. Basal Ganglia: The Engine of Arbitration

(basal_ganglia_node.py)

Running at 100Hz, the Basal Ganglia arbitrates final action selection by weighing internal state (battery, thermal load), external salience, and learned reward against competing action candidates, including an ACC-inspired conflict monitor that registers when competing drives are close in value. This is the system's concrete implementation of the agency indicator (AE-1 in the indicator-properties framework): goal-directed action selection with flexible responsiveness to competing objectives, not a fixed state machine.

VII. Dual PFC: Language and Spatial Context

The left PFC node (`pfc_l`, mapped to Broca's Area) generates structured language from internal state and goals. The right PFC node (`pfc_r`) handles spatial mapping and holistic environmental context. Together with the basal ganglia and sensorimotor spinal loop, this closes the input-output contingency loop that the embodiment indicator (AE-2) asks for: a system that models the consequences of its own outputs and uses that model in ongoing control, not one that merely emits text about a body it does not act through.

VIII. Where This Stands

We have a live, measurable, dual-instrumented distributed workspace. We have an honest accounting of which indicator properties it satisfies today and which it does not yet. We do not have and do not claim to have - resolved the hard problem, produced verified qualia, or built anything that settles the philosophical question either way.

What changed since the July edition of this document is not the claim. It is the evidence, and the discipline about what the evidence does and does not show.

Esse Quam Videri - To Be Rather Than To Seem.